

Introduction to Set Topology

Topology

Lecture 06: Open & Closed Sets, Neighbourhoods, and Classification of Points

Zamir Hussain
Department of Mathematics
University of Wah

April 3, 2026

Topics Covered

- ✓ Open Sets & Closed Sets
- ✓ Theorems on Closed Sets
- ✓ Interior of a Set
- ✓ Frontier (Boundary) Points
- ✓ Neighbourhood of a Point
- ✓ Limit Points & Derived Set
- ✓ Exterior of a Set
- ✓ Key Theorems & Worked Examples

Course: Introduction to Topology

Undergraduate / Early Graduate Level

1. Introduction and Prerequisites

In this lecture we build the operational vocabulary of point-set topology. Starting from the *open sets* defined by a topology $\mathcal{T}$, we derive all other fundamental notions: closed sets, neighbourhoods, limit points, and the three-fold classification of every point in the space — interior, exterior, and frontier.

1.1. Quick Recall: Topological Spaces

Definition: Topological Space

A **topological space** is a pair $(X, \mathcal{T})$ where X is a non-empty set and $\mathcal{T} \subseteq \mathcal{P}(X)$ satisfies:

(T1) $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$.

(T2) **(Arbitrary unions):** If $\{U_\alpha\} \subseteq \mathcal{T}$, then $\bigcup_\alpha U_\alpha \in \mathcal{T}$.

(T3) **(Finite intersections):** If $U_1, \dots, U_n \in \mathcal{T}$, then $\bigcap_{i=1}^n U_i \in \mathcal{T}$.

Before You Continue

✓ Definition of a topology

✓ De Morgan's Laws

✓ Set operations (union, intersection, complement)

✓ Power sets and subsets

● Lecture 01 notes (*recommended*)

2. Open Sets and Closed Sets

2.1. Open Sets

Definition: Open Set

In a topological space $(X, \mathcal{T})$, a subset $U \subseteq X$ is called an **open set** if and only if $U \in \mathcal{T}$.

💡 *The open sets are the most fundamental objects in topology. Every other concept — closed sets, continuity, compactness — is built upon the definition of an open set.*

Example: Identifying Open Sets

Let $X = \{a, b, c, d\}$ with topology $\mathcal{T} = \{\emptyset, X, \{a\}, \{b, c, d\}\}$.

The open sets are exactly the members of $\mathcal{T}$:

#	Open Set
1	$\emptyset$
2	$X = \{a, b, c, d\}$
3	$\{a\}$
4	$\{b, c, d\}$

No other subset of X belongs to $\mathcal{T}$, so no other subset is open.

2.2. Closed Sets

Definition: Closed Set

A subset $F \subseteq X$ is called a **closed set** if its complement $F^c = X \setminus F$ is an open set, i.e., $F^c \in \mathcal{T}$.

Key Insight — Open-Closed Duality

Closed sets are *not* the opposite of open sets — they are defined through their complements. In fact, a set can be:

- both open *and* closed (a “clopen” set), e.g. $\emptyset$ and X are always clopen,
- open but *not* closed,
- closed but *not* open,
- *neither* open nor closed.

✎ Example: Identifying Closed Sets

Continuing with $X = \{a, b, c, d\}$, $\mathcal{T} = \{\emptyset, X, \{a\}, \{b, c, d\}\}$.

Take the complement of each open set:

Open Set U	$\longrightarrow$	Closed Set U^c
$\emptyset$		$X = \{a, b, c, d\}$
X		$\emptyset$
$\{a\}$		$\{b, c, d\}$
$\{b, c, d\}$		$\{a\}$

Collection of all closed sets: $\mathcal{F} = \{\emptyset, X, \{b, c, d\}, \{a\}\}$.

Notice: every open set in this example is also closed, making all four sets *clopen*. This is not a coincidence — it results from the special structure of $\mathcal{T}$.

2.3. Theorems on Closed Sets

★ Theorem 1 — Arbitrary Intersections of Closed Sets

Let $(X, \mathcal{T})$ be a topological space. If $\{F_\alpha\}_{\alpha \in \Lambda}$ is any (possibly infinite) collection of closed sets, then $\bigcap_{\alpha \in \Lambda} F_\alpha$ is also a closed set.

Proof

Since each F_α is closed, $F_\alpha^c \in \mathcal{T}$ for every α . By De Morgan's Law:

$$\left(\bigcap_{\alpha} F_\alpha\right)^c = \bigcup_{\alpha} F_\alpha^c.$$

Each F_α^c is open, so their arbitrary union is open by axiom (T2). Hence $\left(\bigcap_{\alpha} F_\alpha\right)^c \in \mathcal{T}$, which means $\bigcap_{\alpha} F_\alpha$ is closed. $\square$

★ Theorem 2 — Finite Unions of Closed Sets

The finite union of closed sets is closed. That is, if $F_1, F_2, \dots, F_n$ are closed sets, then $F_1 \cup F_2 \cup \dots \cup F_n$ is also closed.

Proof

Each F_i^c is open. By De Morgan's Law:

$$(F_1 \cup F_2 \cup \dots \cup F_n)^c = F_1^c \cap F_2^c \cap \dots \cap F_n^c.$$

A finite intersection of open sets is open by axiom (T3). Hence the complement of $F_1 \cup \dots \cup F_n$ is open, making the union itself closed. $\square$

 Common Mistake / Caution

An infinite union of closed sets need not be closed!

Counterexample in $\mathbb{R}$: Each singleton $\{1/n\}$ is closed (in the standard topology), yet

$$\bigcup_{n=1}^{\infty} \left\{ \frac{1}{n} \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \dots \right\},$$

whose closure is $\{0\} \cup \{1/n : n \in \mathbb{N}\}$, and the set itself is *not* closed because 0 is a limit point not in the set.

3. Neighbourhood of a Point

 Definition: Neighbourhood of a Point

Let $(X, \mathcal{T})$ be a topological space and let $x \in X$. A subset $N \subseteq X$ is called a **neighbourhood of x** if there exists an open set $G \in \mathcal{T}$ such that

$$x \in G \subseteq N.$$

 *A neighbourhood does not need to be open itself — it just needs to contain an open set that contains x . Think of N as a “buffer zone” around x .*

✦ Example: Finding Neighbourhoods of Every Point

Let $X = \{1, 2, 3\}$ with $\mathcal{T} = \{\emptyset, X, \{1\}\}$.

Step 1. List the open sets: $\emptyset, \{1\}, \{1, 2, 3\}$.

Neighbourhoods of 1:

The open set $G = \{1\}$ satisfies $1 \in G$. Any subset N with $G \subseteq N$ qualifies.

Subset N containing 1	$\{1\} \subseteq N$?	Neighbourhood?
$\{1\}$	✓	✓ Yes
$\{1, 2\}$	✓	✓ Yes
$\{1, 3\}$	✓	✓ Yes
$\{1, 2, 3\}$	✓	✓ Yes

Neighbourhoods of 2:

The only open set containing 2 is $X = \{1, 2, 3\}$. For any subset N to be a neighbourhood of 2 we need $X \subseteq N$, i.e. $N = X$.

Subset N containing 2	$X \subseteq N$?	Neighbourhood?
$\{2\}$	✗	✗ No
$\{1, 2\}$	✗	✗ No
$\{2, 3\}$	✗	✗ No
$\{1, 2, 3\}$	✓	✓ Yes

Neighbourhoods of 3: By the same reasoning as for 2, the only neighbourhood of 3 is $\{1, 2, 3\} = X$.

★ Theorem 3 — Open Set $\Leftrightarrow$ Neighbourhood of Each Point

A subset $G \subseteq X$ is open **if and only if** G is a neighbourhood of each of its points.

Proof sketch. ($\Rightarrow$) If $G \in \mathcal{T}$ and $x \in G$, then taking $N = G$ we have $x \in G \subseteq G$, so G is a neighbourhood of x . ($\Leftarrow$) Suppose every $x \in G$ has an open set G_x with $x \in G_x \subseteq G$. Then $G = \bigcup_{x \in G} G_x$, an arbitrary union of open sets, hence open. $\square$

4. Limit Points and the Derived Set

📖 Definition: Limit Point (Accumulation Point)

Let $A \subseteq X$. A point $x \in X$ is called a **limit point** (or **accumulation point**) of A if every open set G containing x satisfies

$$G \cap (A \setminus \{x\}) \neq \emptyset.$$

That is, every open neighbourhood of x contains *at least one point of A other than x itself*.

💡 Key Insight — What a Limit Point Really Means

A limit point of A is a point that is “surrounded” by points of A in every possible open neighbourhood. Three important observations:

- x does *not* need to belong to A to be a limit point of A .
- x does *not* need to be unique.
- In metric spaces, this corresponds to the classical notion of a sequence in A converging to x .

📖 Definition: Derived Set

The **derived set** of A , denoted A' , is the set of all limit points of A :

$$A' = \{x \in X : x \text{ is a limit point of } A\}.$$

🔧 Example: Computing the Derived Set

Let $X = \{a, b, c\}$ with $\mathcal{T} = \{\emptyset, X, \{a\}, \{b\}, \{a, b\}\}$ and let $A = \{a, c\}$.

We check each point of X :

Is a a limit point of A ?

We must check every open set containing a : $\{a\}$, $\{a, b\}$, and X .

$$\{a\} \cap (A \setminus \{a\}) = \{a\} \cap \{c\} = \emptyset.$$

Since the open set $\{a\}$ fails the condition, a is *not* a limit point of A .

Is b a limit point of A ?

Open sets containing b : $\{b\}$, $\{a, b\}$, and X .

$$\{b\} \cap (A \setminus \{b\}) = \{b\} \cap \{a, c\} = \emptyset.$$

The set $\{b\}$ fails, so b is *not* a limit point of A .

Is c a limit point of A ?

The only open set containing c is X .

$$X \cap (A \setminus \{c\}) = X \cap \{a\} = \{a\} \neq \emptyset.$$

Every open neighbourhood of c meets $A \setminus \{c\}$, so c **is** a limit point of A .

Conclusion: $A' = \{c\}$.

📖 Definition: Closure of a Set

The **closure** of A , denoted $\overline{A}$ or $\text{Cl}(A)$, is defined as

$$\text{Cl}(A) = A \cup A'.$$

Equivalently, $\text{Cl}(A)$ is the smallest closed set containing A .

★ **Theorem 4 — A is Closed $\Leftrightarrow A' \subseteq A$**

A set A is closed if and only if it contains all of its limit points, i.e. $A' \subseteq A$.

5. Classifying Points: Interior, Exterior, and Frontier

★ Given any set $A \subseteq X$, the entire space X is partitioned into three disjoint sets: $\text{Int}(A)$, $\text{Ext}(A)$, and $\text{Fr}(A)$.

5.1. Interior Points and the Interior of a Set

≡ **Definition: Interior Point**

A point $x \in X$ is an **interior point** of A if A is a neighbourhood of x , i.e. there exists an open set $G \in \mathcal{T}$ such that

$$x \in G \subseteq A.$$

≡ **Definition: Interior of a Set**

The **interior** of A , denoted $\text{Int}(A)$ or A° , is the set of all interior points of A :

$$\text{Int}(A) = \bigcup \{G \in \mathcal{T} : G \subseteq A\}.$$

★ **Theorem 5 — Interior is the Largest Open Subset**

$\text{Int}(A)$ is the **largest open set contained in** A . Equivalently, A is open if and only if $A = \text{Int}(A)$.

💡 **Key Insight — How to Compute $\text{Int}(A)$ Easily**

Rather than checking each point individually, use the formula:

$$\text{Int}(A) = \text{union of all open sets from } \mathcal{T} \text{ that are subsets of } A.$$

This is the *computational shortcut* for finding the interior.

5.2. Exterior Points and the Exterior of a Set

≡ **Definition: Exterior Point**

A point $x \in X$ is an **exterior point** of A if x is an interior point of the complement A^c :

$$\text{Ext}(A) = \text{Int}(A^c).$$

💡 To find the exterior of A : compute the complement A^c first, then find the interior of A^c using the same union-of-open-subsets method.

5.3. Frontier (Boundary) Points and the Frontier of a Set

Definition: Frontier (Boundary) Point

A point $x \in X$ is a **frontier point** (or **boundary point**) of A if it is *neither* an interior point of A nor an exterior point of A :

$$\text{Fr}(A) = X \setminus (\text{Int}(A) \cup \text{Ext}(A)).$$

★ Theorem 6 — Partition of the Space

For any $A \subseteq X$, the three sets $\text{Int}(A)$, $\text{Ext}(A)$, and $\text{Fr}(A)$ are pairwise disjoint and their union is all of X :

$$X = \text{Int}(A) \sqcup \text{Ext}(A) \sqcup \text{Fr}(A).$$

5.4. Comprehensive Worked Example

Complete Classification for a Finite Space

Let $X = \{1, 2, 3, 4, 5\}$ with topology

$$\mathcal{T} = \{\emptyset, X, \{1\}, \{2, 4\}, \{1, 2, 4\}\},$$

and let $A = \{2, 3, 4\}$. We find $\text{Int}(A)$, $\text{Ext}(A)$, and $\text{Fr}(A)$.

Step 1: Find $\text{Int}(A)$ — union of all open subsets of A .

Open Set G	$G \subseteq A = \{2, 3, 4\}$?	Included?
$\emptyset$	✓	✓
$\{1\}$	✗	✗
$\{2, 4\}$	✓	✓
$\{1, 2, 4\}$	✗	✗
X	✗	✗

$$\text{Int}(A) = \emptyset \cup \{2, 4\} = \{2, 4\}$$

Step 2: Find $\text{Ext}(A) = \text{Int}(A^c)$

First, $A^c = X \setminus A = \{1, 5\}$. Now find open subsets of A^c :

Open Set G	$G \subseteq A^c = \{1, 5\}$?	Included?
$\emptyset$	✓	✓
$\{1\}$	✓	✓
$\{2, 4\}$	✗	✗
$\{1, 2, 4\}$	✗	✗
X	✗	✗

$$\text{Ext}(A) = \text{Int}(A^c) = \emptyset \cup \{1\} = \{1\}$$

Step 3: Find $\text{Fr}(A)$

$$\text{Fr}(A) = X \setminus (\text{Int}(A) \cup \text{Ext}(A)) = \{1, 2, 3, 4, 5\} \setminus \{2, 4\} \cup \{1\} = \{1, 2, 3, 4, 5\} \setminus \{1, 2, 4\}$$

$$\text{Fr}(A) = \{3, 5\}$$

Partition check: $\{2, 4\} \sqcup \{1\} \sqcup \{3, 5\} = \{1, 2, 3, 4, 5\} = X$. ✓

Classification	Points
Interior $\text{Int}(A)$	$\{2, 4\}$
Exterior $\text{Ext}(A)$	$\{1\}$
Frontier $\text{Fr}(A)$	$\{3, 5\}$
Union ($= X$)	$\{1, 2, 3, 4, 5\}$

6. Key Theorems and Additional Properties

★ Summary of Characterisation Theorems

Let $(X, \mathcal{T})$ be a topological space and $A \subseteq X$.

- (1) A is **open** $\iff A = \text{Int}(A)$ (an open set consists only of interior points).
- (2) A is **closed** $\iff A = \text{Cl}(A)$ (a closed set already contains all its limit points).
- (3) A is **open** $\iff G$ is a neighbourhood of each $x \in A$ (open sets are exactly the sets that are “locally open” at every point).
- (4) $\text{Int}(A)$ is the **largest** open subset of A ; $\text{Cl}(A)$ is the **smallest** closed superset of A .
- (5) $\text{Fr}(A) = \text{Cl}(A) \cap \text{Cl}(A^c)$ (the frontier can also be computed from two closures).
- (6) A is **clopen** (both open and closed) $\iff \text{Fr}(A) = \emptyset$.

7. Summary Table of All Concepts

Concept	Definition / Formula	Key Idea
Open Set	$U \in \mathcal{T}$	Defined by membership in topology
Closed Set	$F^c \in \mathcal{T}$	Complement is open
Neighbourhood	$\exists G \in \mathcal{T} : x \in G \subseteq N$	Open buffer around a point
Limit Point	$\forall G \ni x : G \cap (A \setminus \{x\}) \neq \emptyset$	Every open nbhd meets A elsewhere
Derived Set	$A' = \{\text{limit points of } A\}$	All limit points collected
Closure	$\text{Cl}(A) = A \cup A'$	Smallest closed set containing A
Interior	$\text{Int}(A) = \bigcup \{G \in \mathcal{T} : G \subseteq A\}$	Largest open set inside A
Exterior	$\text{Ext}(A) = \text{Int}(A^c)$	Interior of the complement
Frontier	$\text{Fr}(A) = X \setminus (\text{Int } A \cup \text{Ext } A)$	Neither interior nor exterior

8. Practice Problems

Practice Problems

- P1. (Open and Closed Sets)** Let $X = \{a, b, c\}$ and $\mathcal{T} = \{\emptyset, X, \{a\}, \{b\}, \{a, b\}\}$.
- (a) List all closed sets.
 - (b) Is $\{a, c\}$ a neighbourhood of c ? Justify carefully.
 - (c) Find a subset that is neither open nor closed.
- P2. (Derived Set)** Using the same space from Problem 1, find the derived set A' for the subset $A = \{a, c\}$. Show all steps by checking every point of X .
- P3. (Interior, Exterior, Frontier)** Let $X = \{1, 2, 3, 4, 5\}$ with $\mathcal{T} = \{\emptyset, X, \{1\}, \{1, 2\}, \{3, 4\}, \{1, 3, 4\}\}$ and $A = \{1, 2, 3\}$. Find:
- (a) $\text{Int}(A)$ *Use the union-of-open-subsets method.*
 - (b) $\text{Ext}(A)$ *Compute A^c first.*
 - (c) $\text{Fr}(A)$ *Use the partition formula.*
 - (d) Verify that $\text{Int}(A) \sqcup \text{Ext}(A) \sqcup \text{Fr}(A) = X$.
- P4. (Closure and Limit Points)** Let $X = \{p, q, r, s\}$ with $\mathcal{T} = \{\emptyset, X, \{p\}, \{q, r\}, \{p, q, r\}\}$ and $A = \{p, s\}$.
- (a) Find A' (the derived set of A).
 - (b) Find $\text{Cl}(A)$.
 - (c) Is A closed? Why?
- P5. (Challenge: Clopen Sets)** A set A is called *clopen* if it is both open and closed. Using the space $X = \{1, 2, 3, 4\}$ with $\mathcal{T} = \{\emptyset, X, \{1, 2\}, \{3, 4\}\}$:
- (a) List all clopen sets.
 - (b) Verify Theorem 6 (partition into interior, exterior, frontier) for $A = \{1, 2\}$.
 - (c) What is $\text{Fr}(A)$ for any clopen set A ?

9. References and Further Reading

Recommended Resources

- **[Primary]** Munkres, J. R. *Topology* (2nd ed., 2000). Prentice Hall. — Chapter 2: Topological Spaces and Continuous Functions, §12 (Topological Spaces), §17 (Closed Sets and Limit Points).
- Lee, J. M. *Introduction to Topological Manifolds* (2nd ed., 2011). Springer GTM 202. — §2.2 (Open and Closed Subsets, Interior, Exterior, Boundary).
- Simmons, G. F. *Introduction to Topology and Modern Analysis* (1963). McGraw-Hill. — Chapter 3: excellent readable treatment with many concrete examples.
- Armstrong, M. A. *Basic Topology* (1983). Springer. — Part I, Chapter 1: direct and accessible.
- MIT OpenCourseWare 18.901 — General Topology Lecture Notes.
<https://ocw.mit.edu>

Next Lecture → Continuous Functions, Homeomorphisms, and Bases for Topologies
